AIMS
Airbus Material Specification

Aluminium alloy (7010 / 7050)
Solution treated, controlled stretched and artificially aged (T7451)
Plate
50,0 mm $\leq$ a $\leq$ 200,0 mm
Close tolerance flatness

Material Specification

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Contents

1 Scope

This Material Specification defines the requirements of aluminium alloy 7010/7050 T7451plate 50,0 ≤ a ≤ 200,0 mm for aerospace application.
Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

No particular application

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 03-02-000	Airbus Material Specification – Technical specification, Aluminium and aluminium alloy wrought products – Plate
ABS 5323	Aerospace series – Plates – Aluminium alloy 7010 / 7050 close tolerance flatness Thickness 6,0 mm ≤ a ≤ 200,0 mm, Dimensions
EN 2032-2	Aerospace series – Metallic materials – Part 2: Coding of the metallurgical condition in delivery condition
EN 4500-2	Aerospace series – Metallic materials – Rules for drafting and presentation of material standards, Part 2: Specific rules for aluminium, aluminium alloys and magnesium alloys
EN 6072	Aerospace series – Metallic materials – Test methods – Constant amplitude fatigue testing
ASTM G 34	Standard Test Method for Exfoliation Corrosion Susceptibility in 2xxx and 7xxx Series Aluminum Alloys (EXCO Test)

4 Technical requirements

4.1 General

Specification values are minimum except where stated.

4.2 Specific

See tables according to EN 4500-2

5 Designation

Refer to ABS 5323

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 03-02-000.

1	Material designation		Aluminium alloy 7010 / 7050											
2	Chemical composition %	Element	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Zr	others each	others total	Al
		min.	see relevant IPS											Base
		max.												
3	Method of melting		-											
4	Form Method of production Limit dimension (mm)		Plate Rolled 50 ≤ a ≤ 200											
5	Technical specification		AIMS 03-02-000											
6	6.1 Delivery condition and heat treatment		T7451 Solution treated 470°C ≤ θ ≤ 480°C / WQ θ ≤ 40°C + controlled stretched (1,5 – 3 %) + two-step artificially aged 115°C ≤ θ ≤ 125°C / 3h ≤ t ≤ 14h 175°C ≤ θ ≤ 185°C / 6h ≤ t ≤ 7h **or** 155°C ≤ θ ≤ 170°C / 20h ≤ t ≤ 30h											
	6.2 Delivery condition code		U [1)]											
7	Use condition and heat treatment		T7451 as delivered											
8	Sample test piece heat treatment		T7451											

9		Dimension concerned	a	mm	50 ≤ a ≤ 75			75 < a ≤ 100			100 < a ≤ 125		
10		Thickness of cladding on each face		%	-								
11		Direction of test piece			L	LT	ST	L	LT	ST	L	LT	ST
12	TENSILE	Temperature	θ	°C	Room temperature								
13	TENSILE	Proof stress	$R_{p0,2}$	MPa	435		405	425		400	420		395
14	TENSILE	Strength	R_m	MPa	505		470	495		470	490		465
15	TENSILE	Elongation	A	%	8	7	3	8	5	3	8	5	3
16	TENSILE	Reduction of area	Z	%	-								

17		Hardness	HB	-	-
18		Shear strength	R_c	MPa	-
19		Bending	k	-	-
20		Impact strength	k	-	-
21	CREEP	Temperature	θ	°C	-
22	CREEP	Time		h	-
23	CREEP	Stress	σ_a	MPa	-
24	CREEP	Elongation	a	%	-
25	CREEP	Rupture stress	σ_R	MPa	-
26	CREEP	Elongation at rupture	A	%	-
27		Notes (see Line 98)			1) and 2)

1	Material designation		Aluminium alloy 7010 / 7050											
2	Chemical composition %	Element	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Zr	others each	others total	Al
		min.	see relevant IPS											Base
		max.												
3	Method of melting		-											
4	Form Method of production Limit dimension (mm)		Plate Rolled $50 \leq a \leq 200$											
5	Technical specification		AIMS 03-02-000											
6	6.1 Delivery condition and heat treatment		T7451 Solution treated 470°C ≤ θ ≤ 480°C / WQ θ ≤ 40°C + controlled stretched (1,5 – 3 %) + two-step artificially aged 115°C ≤ θ ≤ 125°C / 3h ≤ t ≤ 14h 175°C ≤ θ ≤ 185°C / 6h ≤ t ≤ 7h **or** 155°C ≤ θ ≤ 170°C / 20h ≤ t ≤ 30h											
	6.2 Delivery condition code		U 1)											
7	Use condition and heat treatment		T7451 as delivered											
8	Sample test piece heat treatment		T7451											

9	Dimension concerned		a	mm	$125 < a \leq 150$			$150 < a \leq 175$			$175 < a \leq 200$		
10	Thickness of cladding on each face			%	-								
11	Direction of test piece				L	LT	ST	L	LT	ST	L	LT	ST
12	TENSILE	Temperature	θ	°C	Room temperature								
13	TENSILE	Proof stress	$R_{p0,2}$	MPa	415		395	405		390	405		385
14	TENSILE	Strength	R_m	MPa	485		460	475		455	470		450
15	TENSILE	Elongation	A	%	7	4	3	6	3	3	5	3	3
16	TENSILE	Reduction of area	Z	%	-								

17	Hardness		HB	-	-
18	Shear strength		R_c	MPa	-
19	Bending		k	-	-
20	Impact strength		k	-	-
21	CREEP	Temperature	θ	°C	-
22	CREEP	Time		h	-
23	CREEP	Stress	σ_a	MPa	-
24	CREEP	Elongation	a	%	-
25	CREEP	Rupture stress	σ_R	MPa	-
26	CREEP	Elongation at rupture	A	%	-
27	Notes (see Line 98)				1) and 2)

<table>
<tr><td rowspan="3">32</td><td rowspan="3">Electrical conductivity
γ [MS/m]</td><td rowspan="3">7</td><td>$\gamma \geq 23,0$</td><td colspan="4">Acceptable</td></tr>
<tr><td>$22,0 \leq \gamma < 23,0$</td><td colspan="4">Acceptable if $R_{p0,2\ LT} \leq R_{p0,2\ LT\ min} + 70$ MPa
or stress corrosion test is acceptable</td></tr>
<tr><td>$\gamma < 22,0$</td><td colspan="4">Not acceptable</td></tr>
<tr><td rowspan="2">39</td><td rowspan="2">Stress corrosion</td><td>6</td><td colspan="5">ST direction, tension, constant load, stress level 240 MPa
Length of exposure: 30 days</td></tr>
<tr><td>7</td><td colspan="5">Criterion: no failure</td></tr>
<tr><td rowspan="6">40</td><td rowspan="6">Fracture toughness
K_{IC} [MPa√m]</td><td rowspan="6">7</td><td>Dimension [mm]</td><td>L–T</td><td>T–L</td><td colspan="2">S–L</td></tr>
<tr><td>$50 \leq a \leq 75$</td><td>30</td><td>26</td><td rowspan="5" colspan="2">23</td></tr>
<tr><td>$75 < a \leq 100$</td><td>28</td><td>25</td></tr>
<tr><td>$100 < a \leq 125$</td><td>27</td><td>24</td></tr>
<tr><td>$125 < a \leq 150$</td><td>26</td><td>24</td></tr>
<tr><td>$150 < a \leq 200$</td><td>25</td><td>23</td></tr>
<tr><td>44</td><td>External defects</td><td>–</td><td colspan="5">See AIMS 03-02-000</td></tr>
<tr><td rowspan="15">46</td><td rowspan="15">Fatigue (open hole)

continued</td><td>3</td><td colspan="5">T-type specimen in accordance with EN 6072
$50 \leq a \leq 150$ mm</td></tr>
<tr><td rowspan="14">7</td><td colspan="5">The curve of the test results, calculated in accordance with EN 6072, shall not lie below the reference curve defined by following data:</td></tr>
<tr><td>$\sigma_{max\ nett}$ [MPa]</td><td>Number of cycles</td><td>R</td><td>K_t</td><td>Direction</td></tr>
<tr><td>290</td><td>1 E04</td><td rowspan="5">0,1</td><td rowspan="5">2,3</td><td rowspan="5">T–L</td></tr>
<tr><td>210</td><td>3 E04</td></tr>
<tr><td>155</td><td>1 E05</td></tr>
<tr><td>145</td><td>2 E05</td></tr>
<tr><td>130</td><td>1 E06</td></tr>
<tr><td colspan="5">or</td></tr>
<tr><td>$\sigma_{max\ nett}$ [MPa]</td><td>Number of cycles</td><td>R</td><td>K_t</td><td>Direction</td></tr>
<tr><td>240</td><td>1 E04</td><td rowspan="5">0,1</td><td rowspan="5">2,5</td><td rowspan="5">T–L</td></tr>
<tr><td>225</td><td>2 E04</td></tr>
<tr><td>150</td><td>1 E05</td></tr>
<tr><td>130</td><td>2 E05</td></tr>
<tr><td>105</td><td>1 E06</td></tr>
</table>

<table>
<tr><td rowspan="4">46</td><td rowspan="4">Fatigue (open hole)</td><td>3</td><td>T-type specimen in accordance with EN 6072
150 < a ≤ 200 mm</td></tr>
<tr><td>7</td><td>The curve of the test results, calculated in accordance with EN 6072, shall not lie below the reference curve defined by following data:
<table>
<tr><th>$\sigma_{max\ nett}$ [MPa]</th><th>Number of cycles</th><th>R</th><th>K_t</th><th>Direction</th></tr>
<tr><td>260</td><td>1 E04</td><td rowspan="6">0,1</td><td rowspan="6">2,3</td><td rowspan="6">T–L</td></tr>
<tr><td>210</td><td>2 E04</td></tr>
<tr><td>185</td><td>3 E04</td></tr>
<tr><td>140</td><td>1 E05</td></tr>
<tr><td>120</td><td>2 E05</td></tr>
<tr><td>110</td><td>1 E06</td></tr>
</table></td></tr>
<tr><td>3</td><td>Mini T-type specimen in accordance with EN 6072
150 < a ≤ 200 mm</td></tr>
<tr><td>7</td><td>The curve of the test results, calculated in accordance with EN 6072, shall not lie below the reference curve defined by following data:
<table>
<tr><th>$\sigma_{max\ nett}$ [MPa]</th><th>Number of cycles</th><th>R</th><th>K_t</th><th>Direction</th></tr>
<tr><td>140</td><td>3 E04</td><td rowspan="4">0,1</td><td rowspan="4">2,3</td><td rowspan="4">S–L</td></tr>
<tr><td>105</td><td>1 E05</td></tr>
<tr><td>85</td><td>2 E05</td></tr>
<tr><td>65</td><td>1 E06</td></tr>
</table></td></tr>
<tr><td rowspan="3">46</td><td rowspan="3">Fatigue
(Smooth fatigue test)</td><td>–</td><td>See AIMS 03-02-000</td></tr>
<tr><td>6</td><td>Thickness: 150 ≤ a ≤ 200 mm</td></tr>
<tr><td>7</td><td>The specimens shall be tested with R = 0,1 at a maximum stress of 241 MPa and shall meet the following fatigue requirements:
– Minimum individual life time: 90 000 cycles
– Log average of 4 samples: 120 000 cycles
– Runout: 200 000 cycles</td></tr>
<tr><td>49</td><td>Exfoliation corrosion</td><td>7</td><td>Equal or better grade EB of ASTM G34 [2)]</td></tr>
<tr><td>61</td><td>Internal defects</td><td>–</td><td>See AIMS 03-02-000</td></tr>
<tr><td>69</td><td>Elastic modulus</td><td>7</td><td>The value must be within the following range
69 000 ≤ E ≤ 73 000 MPa
71 000 ≤ E_C ≤ 75 000 MPa</td></tr>
<tr><td>70</td><td>Compression Proof stress
$R_{0,2C}$ [MPa]</td><td>7</td><td>
<table>
<tr><th>Dimension [mm]</th><th>L</th><th>LT</th><th>ST</th></tr>
<tr><td>50 ≤ a ≤ 75</td><td>420</td><td>455</td><td rowspan="3">435</td></tr>
<tr><td>75 < a ≤ 100</td><td>415</td><td>450</td></tr>
<tr><td>100 < a ≤ 125</td><td>400</td><td>440</td></tr>
<tr><td>125 < a ≤ 150</td><td>395</td><td>435</td><td>425</td></tr>
<tr><td>150 < a ≤ 180</td><td>390</td><td>415</td><td>420</td></tr>
<tr><td>180 < a ≤ 200</td><td>385</td><td>410</td><td>415</td></tr>
</table></td></tr>
</table>

71	Crack propagation	3	CCT specimen, W = 160 mm, $2a_i$ = 4,0 mm, $2a_f$ = 53 mm, thickness = 5,0 mm, L–T and T–L direction				
		7	The crack propagation rates shall not be greater than the following data:				
			ΔK [MPa√m]	da/dN [mm/cycle]	R		
			5	2,0 E-5	0,1		
			10	2,0 E-4			
			20	2,0 E-3			
			25	1,5 E-2			
			30	5,0 E-2			
82	Batch uniformity	–	See AIMS 03-02-000				
		7	Electrical conductivity	γ = 24,0 MS/m (typical value) δ ≤ 1,0 MS/m per product Δ ≤ 1,5 MS/m per batch			
			or				
			Hardness	140 HB (typical value) δ ≤ 20 HB per product Δ ≤ 30 HB per batch			
85	Bearing stress F_{bry} and F_{bru} [MPa]	7	Dimension [mm]	F_{bry}		F_{bru}	
				e/D = 1,5	e/D = 2,0	e/D = 1,5	e/D = 2,0
			50 ≤ a ≤ 75	615	715	745	970
			75 < a ≤ 100	620	715	745	965
			100 < a ≤ 125	620	725	740	950
			125 < a ≤ 180	620	725	725	950
			180 < a ≤ 200	610	715	715	930
98	Notes	–	1) See EN 2032-2 2) Limitation: Test solution quantity 15 ± 1 ml/cm^2				
99	Typical use	–					
100	Qualification		Frequency of qualification testing is included in AIMS 03-02-000. Recommended thickness are 63, 90, 110, 140, 170 and 200 mm.				

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 04/2000	–	New standard
2 10/2000	Line 46 Line 71	Fatigue and crack propagation requirements modified
3 05/2002	Line 71 Line 85	Crack propagation and bearing requirements modified
4 07/2004	Line 2 Line 32 Line 71	Chemical composition, electrical conductivity and crack propagation requirements modified
5 06/2005	Line 46	Smooth fatigue test requirements added
6 01/2006	Line 82	Typing error (Δ for electrical conductivity) rectified
7 01/2010	General Paragraph 3 Line 2 Line 6 Line 9 and 82 Line 32 Line 40 Line 85	Editorial changes Normative references updated Zr as alloying element added Stretch rate changed to 3% Typing errors ($125 < a \leq 150$, AIMS 03-02-000) rectified Wording changed to avoid misinterpretation Thickness ranges $150 < a \leq 180$ and $180 < a \leq 200$ merged (same requirements) Thickness ranges $125 < a \leq 150$ and $150 < a \leq 180$ merged (same requirements)